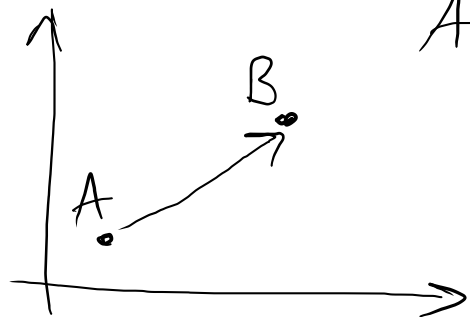
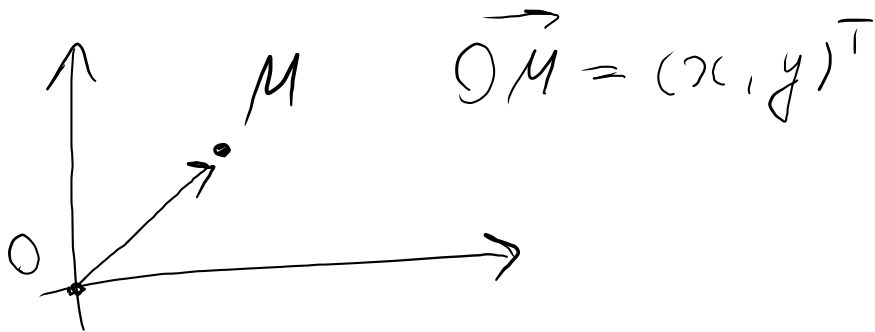


Вычислительная геометрия

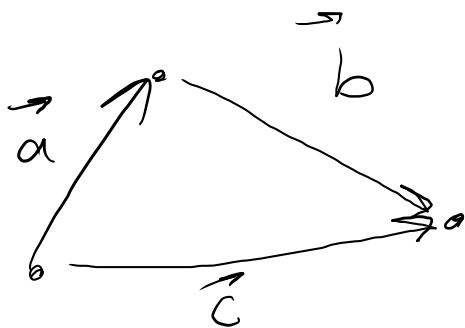


$\overrightarrow{AB}$ - вектор $\overrightarrow{AB} = (x, y)^T$



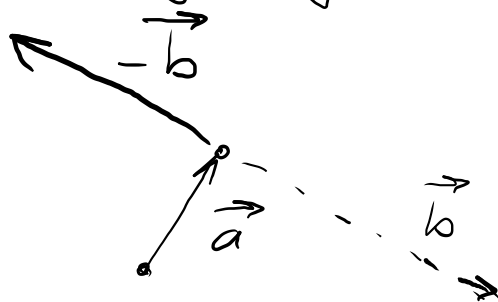
$\overrightarrow{OM} = (x, y)^T$

$$\vec{a} + \vec{b} = \vec{c}$$



$$\vec{c} = (a.x + b.x, a.y + b.y)^T$$

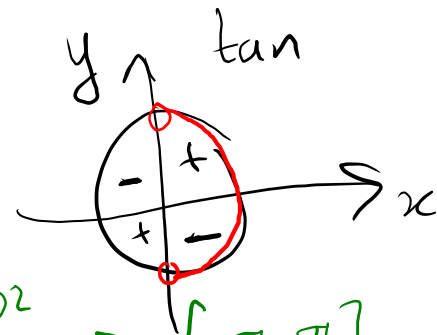
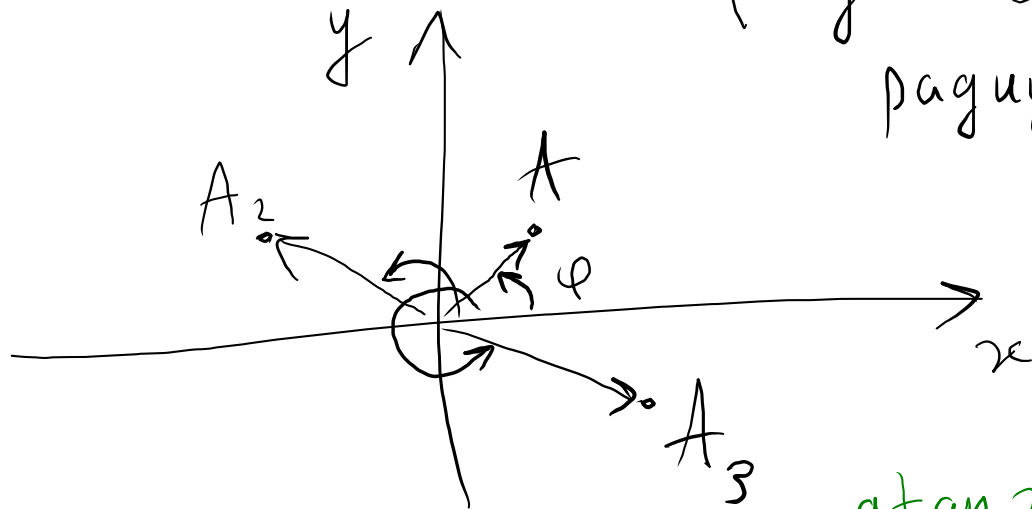
$\vec{a} - \vec{b}$



Поларный график

$$\varphi \in [0; 2\pi)$$

φ - угол от Ox до
радиус-вектора A



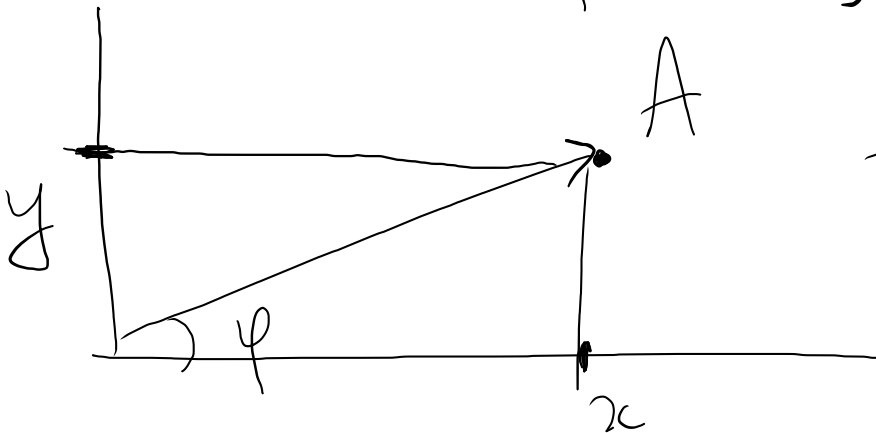
$$\text{atan2} : \mathbb{R}^2 \rightarrow [-\pi, \pi]$$

$$\text{tg } \varphi = \frac{y}{x}$$

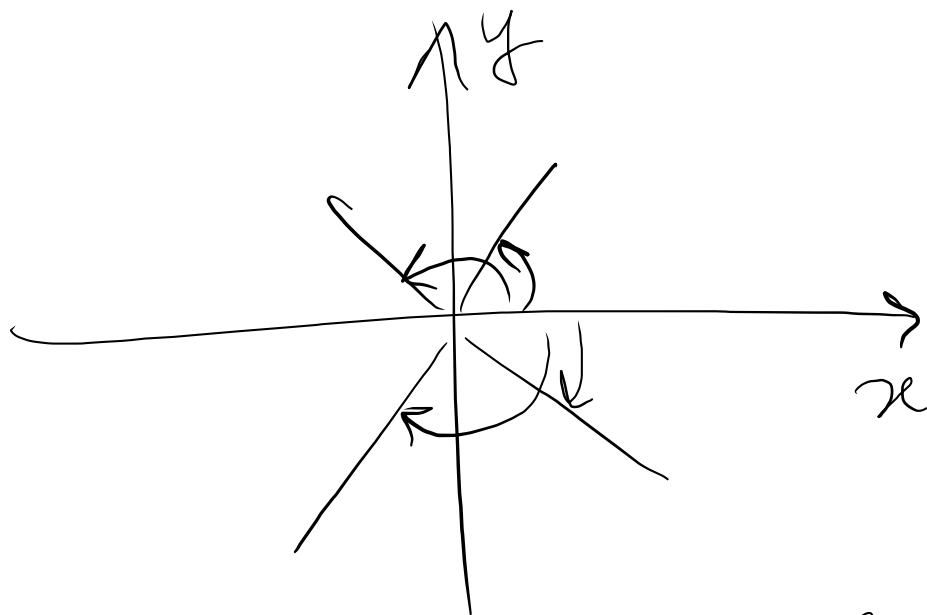
$$\varphi = \text{atan2}(y, x)$$

$$\varphi = \text{atan}\left(\frac{y}{x}\right)$$

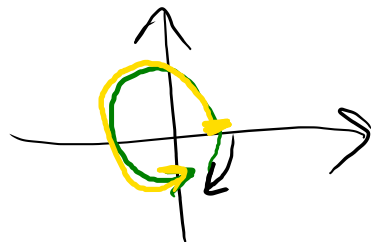
$$\text{atan} : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}; \frac{\pi}{2}\right)$$



$\operatorname{atan} z$



$[-\pi, \pi]$



φ - полярный угол

$[-\pi; \pi] \rightarrow [0; 2\pi)$

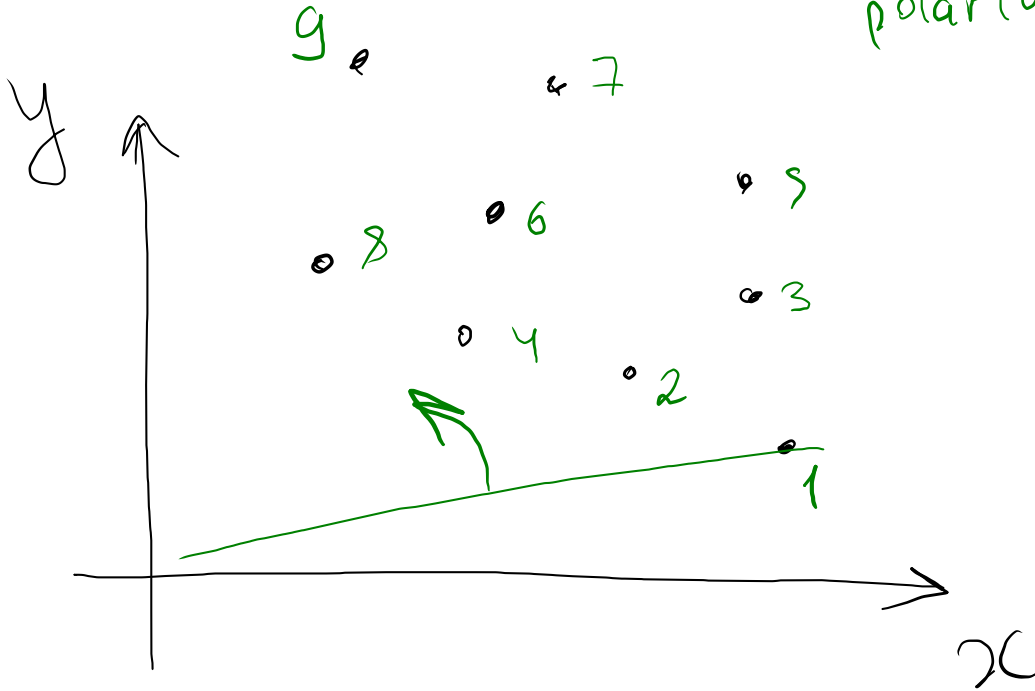
$\varphi \in [0; 2\pi)$

$\operatorname{atan} z < 0$

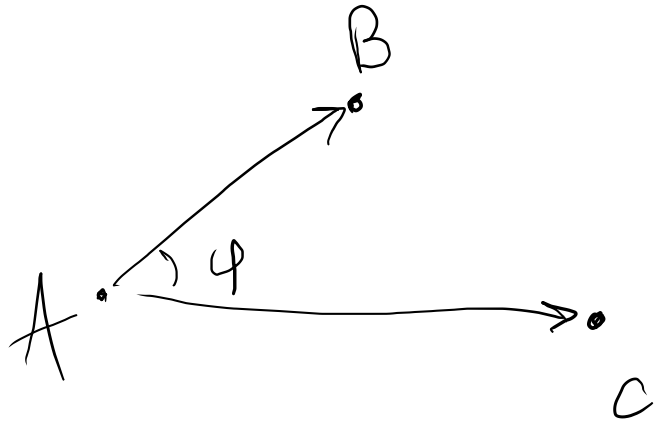
$\varphi = 2\pi + \operatorname{atan} z$

Сортировка по полярному углу

bool cmp(a, b)
 $\text{polar}(a) < \text{polar}(b)$



Угол между векторами



$$\varphi = ?$$

$|\vec{a}|$ — длина $\vec{a}$

$$|\vec{a}| = \sqrt{a_x^2 + a_y^2} = \sqrt{\vec{a} \cdot \vec{a}}$$

$$\vec{a} \cdot \vec{b} = (\vec{a}, \vec{b}) = |\vec{a}| \cdot |\vec{b}| \cdot \cos \varphi = a_x \cdot b_x + a_y \cdot b_y$$

Скалярное произв $\vec{a}$ и $\vec{b}$ dot

$$\vec{a} \times \vec{b} = [\vec{a}, \vec{b}] = \vec{c} \quad |\vec{c}| = |\vec{a}| \cdot |\vec{b}| \cdot \sin \varphi$$

Векторное произв $\vec{a}$ и $\vec{b}$ cross

$$\vec{c} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

$$|\vec{a} \times \vec{b}| = |\vec{c}|$$

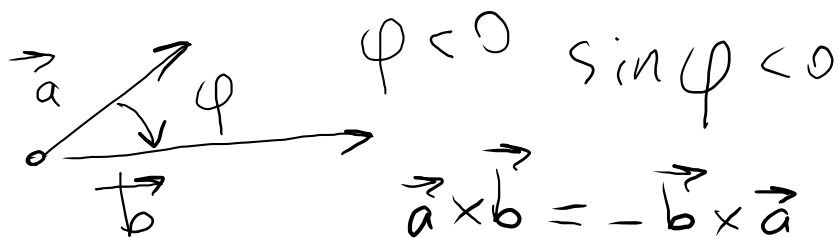
$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a.x & a.y & 0 \\ b.x & b.y & 0 \end{vmatrix} = \vec{k} \cdot (a.x \cdot b.y - a.y \cdot b.x)$$

$$|\vec{c}| = a.x \cdot b.y - a.y \cdot b.x = |\vec{a}| \cdot |\vec{b}| \cdot \sin \varphi$$

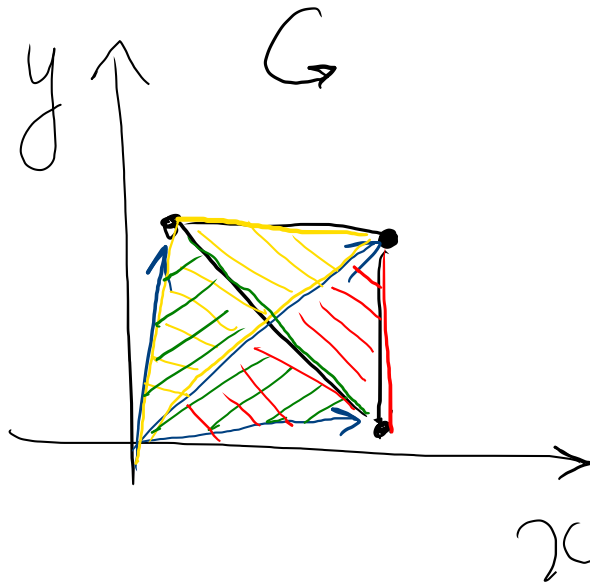
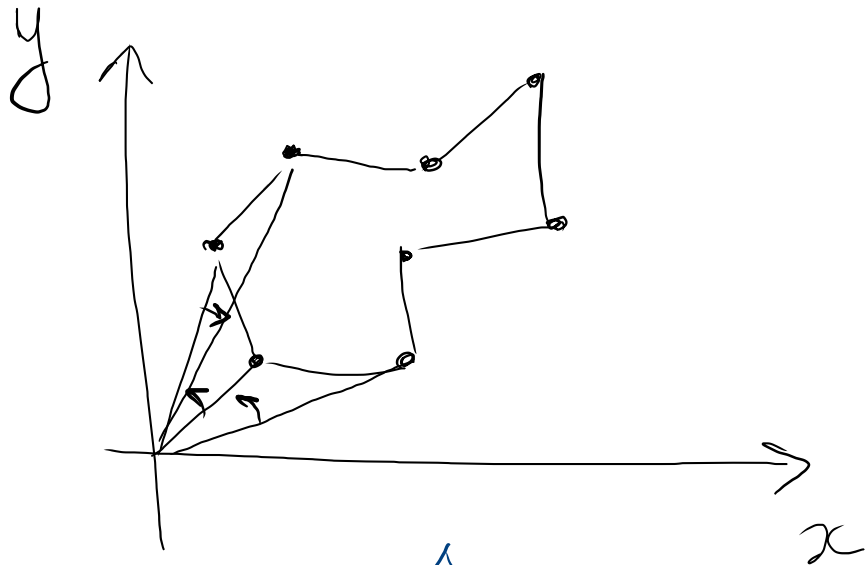
$$\text{atan2}(\overset{x}{?}, \overset{y}{?})$$

$$\tan \varphi = \frac{\sin \varphi}{\cos \varphi} = \frac{|\vec{a}| \cdot |\vec{b}| \sin \varphi}{|\vec{a}| \cdot |\vec{b}| \cos \varphi} = \frac{|\vec{a} \times \vec{b}|}{\vec{a} \cdot \vec{b}}$$

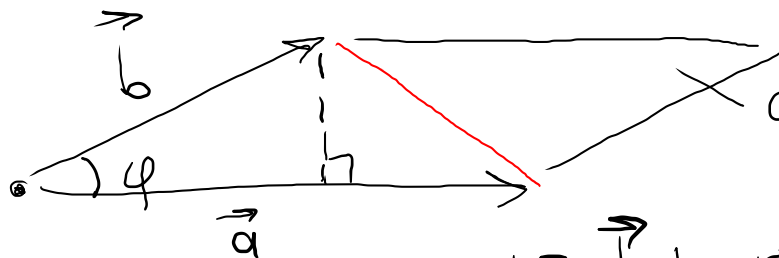
Множественность



$$\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$$

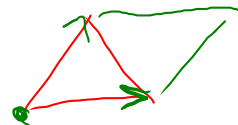


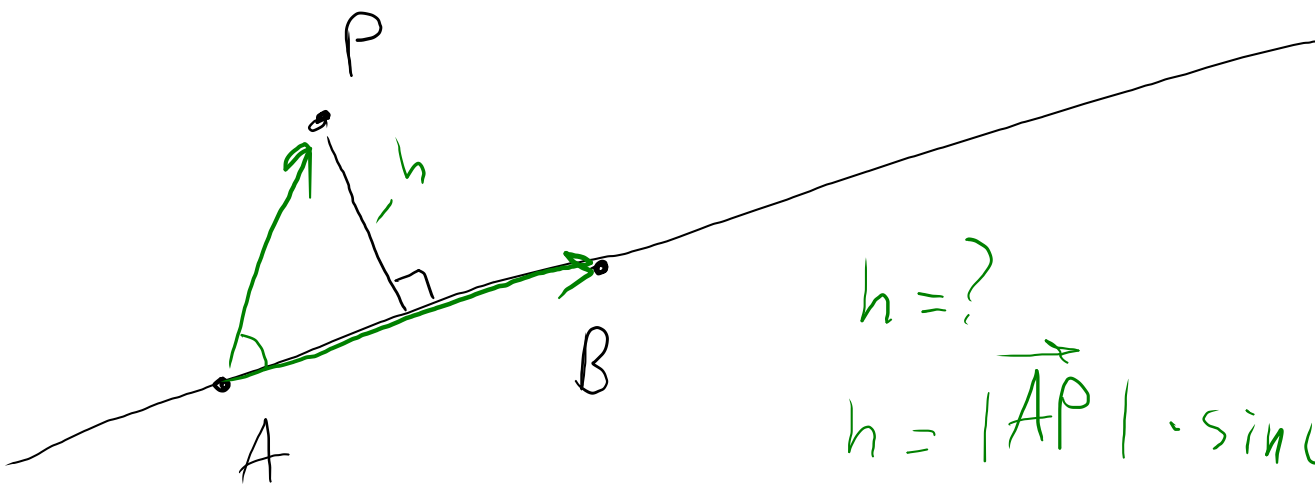
c_{12} - площадь $\triangle$



c - площадь параллелогр.

$$c = |\vec{a} \times \vec{b}| = |\vec{a}| \cdot |\vec{b}| \cdot \sin \varphi$$





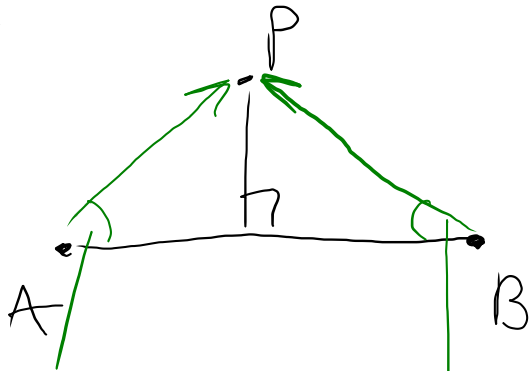
$$h = ?$$

$$h = |\vec{AP}| \cdot \sin \varphi$$

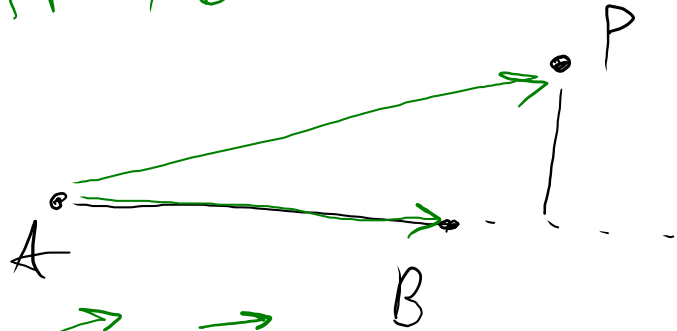
$$|\vec{AP} \times \vec{AB}| = |\vec{AB}| \cdot |\vec{AP}| \cdot \sin \varphi = |\vec{AB}| \cdot h$$

$$h = \frac{|\vec{AP} \times \vec{AB}|}{|\vec{AB}|}$$

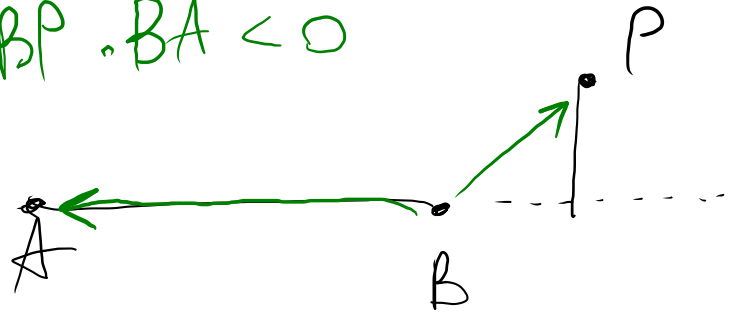
I



$$\vec{AP} \cdot \vec{AB} > 0$$



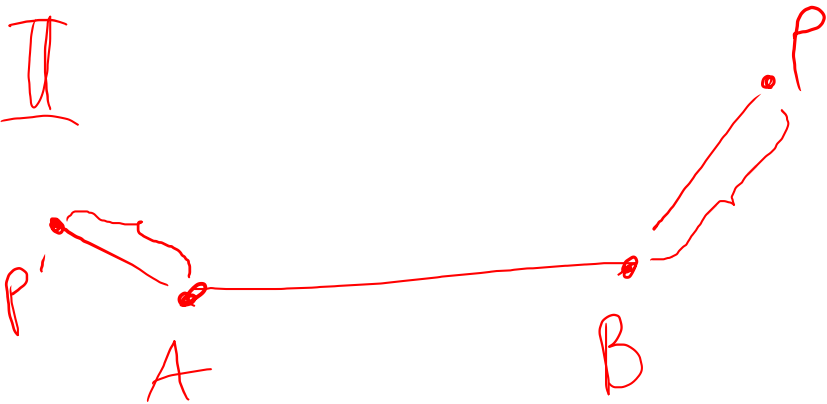
$$\vec{BP} \cdot \vec{BA} < 0$$



$$\vec{AP} \cdot \vec{AB} > 0$$

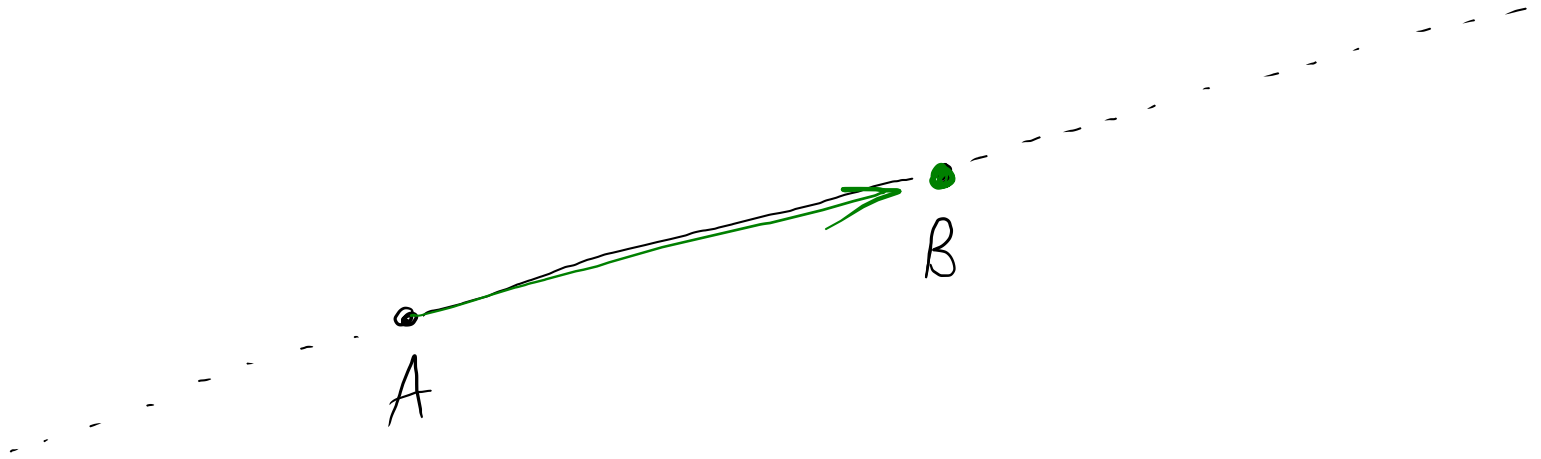
$$\vec{BP} \cdot \vec{BA} > 0$$

II



Принадлежность точки отрезку

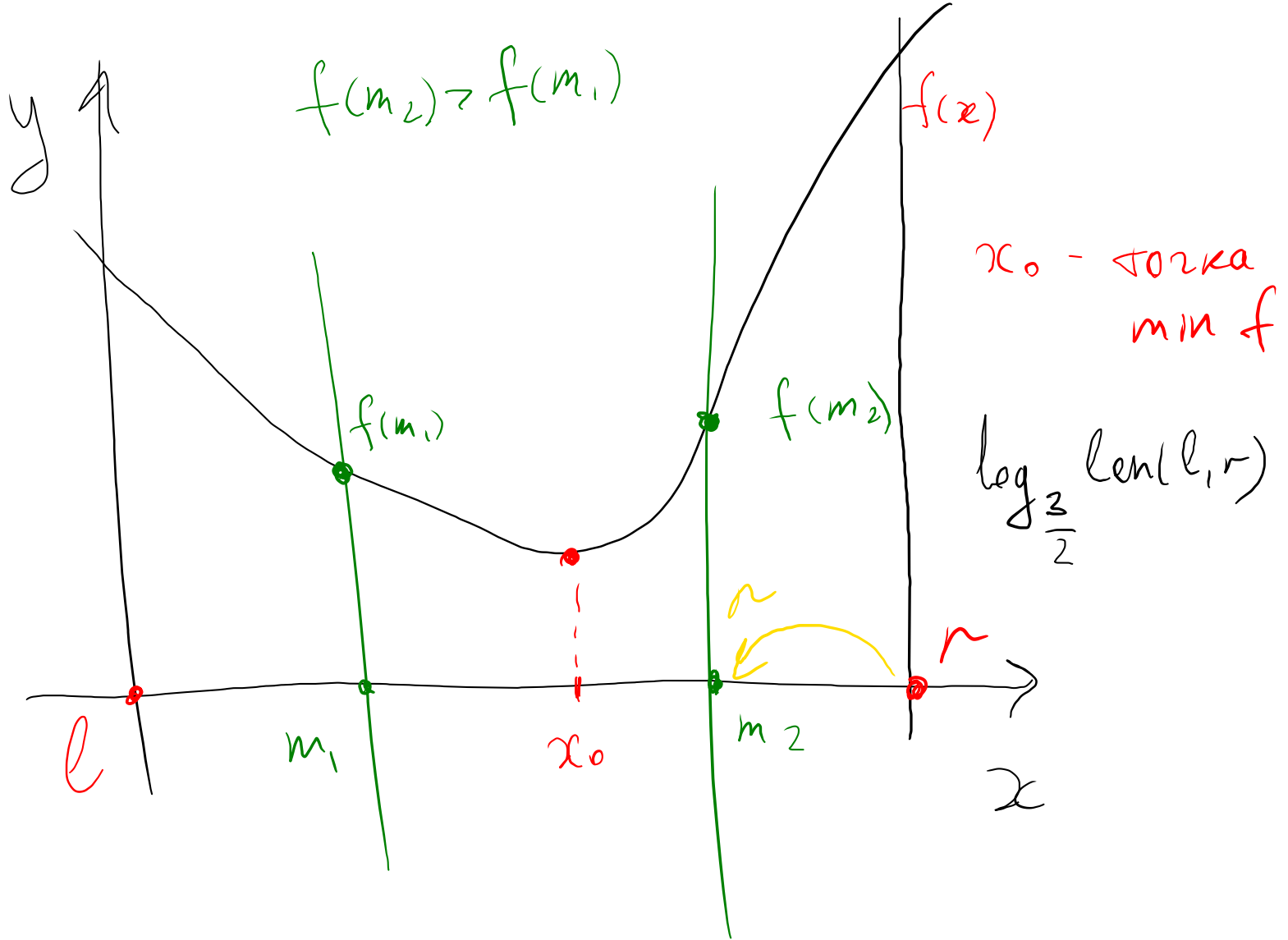
$$Ax + By + C = 0 \quad \text{не точный}$$



$$\vec{AP} \times \vec{AB} = 0 \rightarrow P \text{ лежит на прямой } AB$$

$$0 \leq \vec{AP} \cdot \vec{AB} \leq \vec{AB} \cdot \vec{AB} \rightarrow P \text{ на отрезке } AB$$

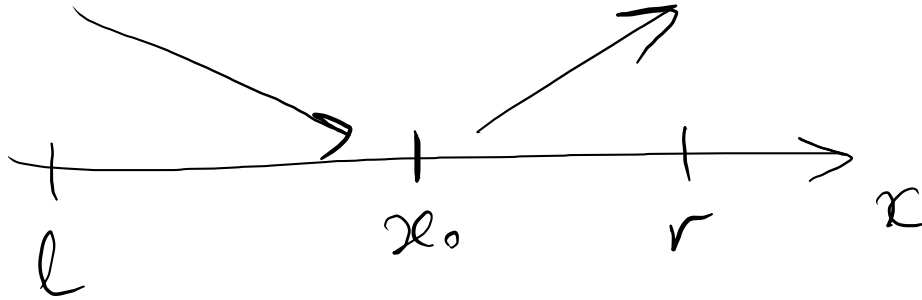
Тернарный поиск



Умногаемость на l, r

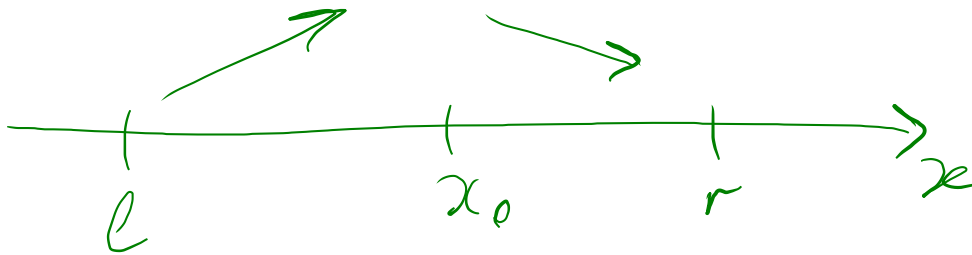
$f(x)$

$f(x_0) - \min$

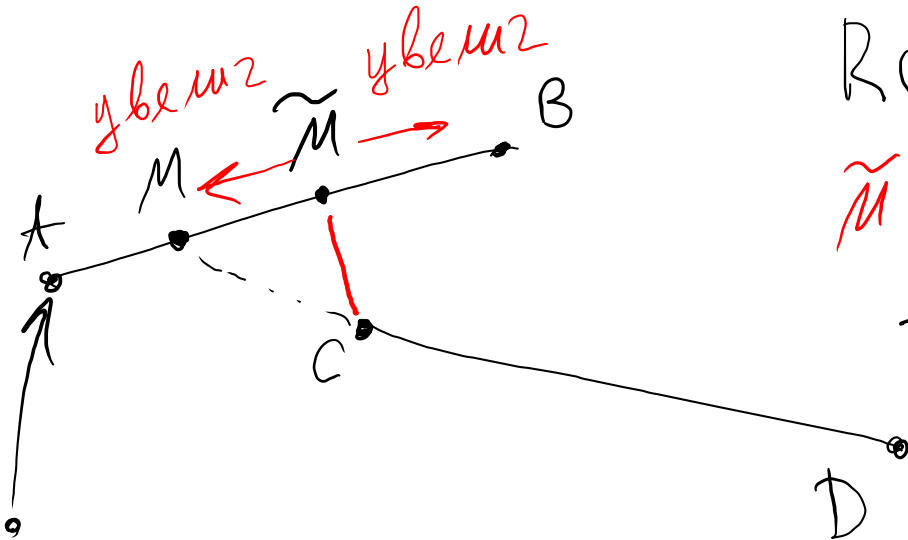
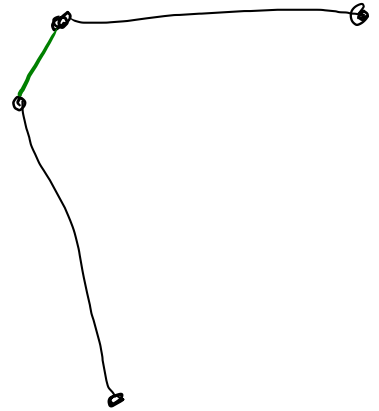
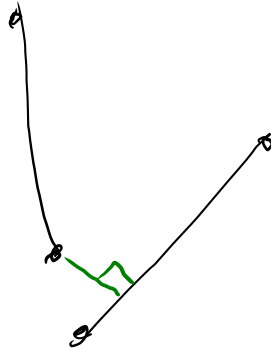
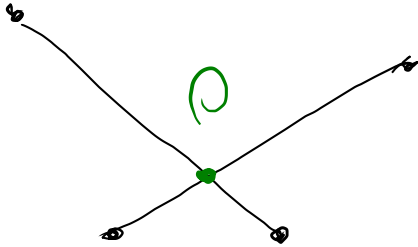


Макс

$f(x_0) - \max$



Расстояние между отрезками



$R(M, CD)$ — путь от M до C

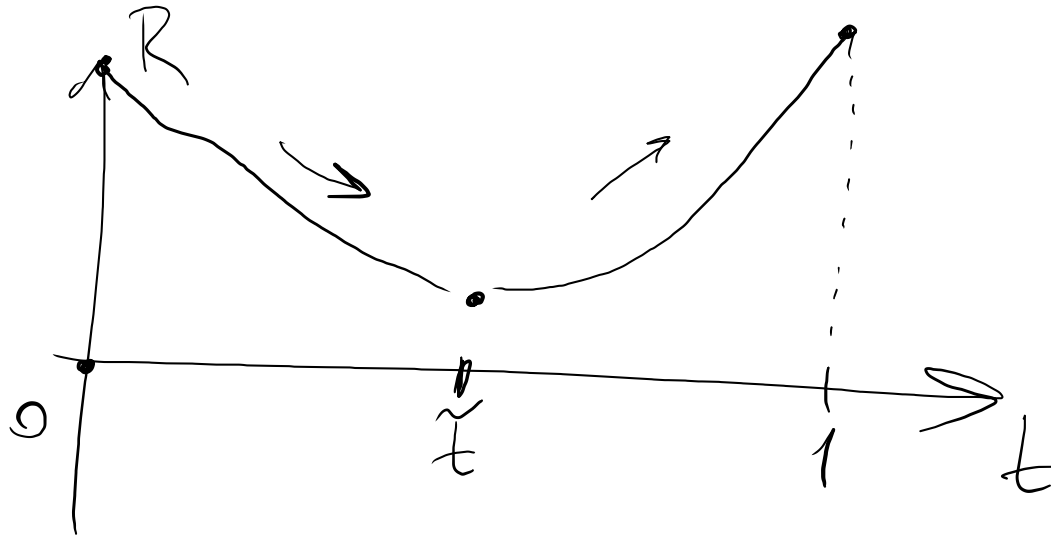
$\tilde{M} \rightarrow R(M, CD) - \min$

$t \in [0, 1]$

$$M(t) = \vec{A} + t \cdot \vec{AB}$$

$$R(M(t), CP) \rightarrow \min$$

$$R(t, CP) \rightarrow \min \quad t \in [0, 1]$$



$R(t)$ — ymu nog.